



## Original Article

# Risk factors for seizure recurrence in children after first unprovoked seizure

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**Abstract** **Background:** The question of whether to treat a patient after a first unprovoked seizure is controversial. This prospective study assessed the time to recurrence and risk factors for seizure recurrence after a first unprovoked seizure in children. **Methods:** Participants were recruited between 1 July 1997, and 30 June 2009. Eligible candidates were children between 1 month and 15 years old who presented with their first unprovoked afebrile seizure. After enrollment, recurrence of seizures was investigated. All participants were followed for at least 2 years. Log-rank test was used for bivariate analysis to check associations, and hazard ratios were used to analyze variables and clinical outcome (recurrence) during follow-up.

**Results:** Of 73 subjects, 42 (57.5%) experienced recurrence. The overall product-limit estimate of recurrence was 61.9% at 6 months, 85.7% at 1 year, and 95.2% at 2 years after seizure onset, respectively. Incidence of recurrence with partial and generalized seizures was 69.0% and 31.0%, respectively. Children with partial seizures had recurrence significantly more often than those with generalized seizures ( $P < 0.001$ ). Recurrent seizures occurred after normal findings on electroencephalogram (EEG) in 21.4%, after generalized spike-and-wave complexes in 16.7%, and after focal epileptic discharge in 61.9%. Children with focal epileptic discharge had recurrence significantly more often than children with normal EEG findings ( $P < 0.001$ ).

**Conclusion:** The time to seizure recurrence after first unprovoked seizure may be within 1 year, and particularly within 6 months; and partial seizure and abnormal EEG with focal epileptic discharge may be risk factors for seizure recurrence.

**Key words** first unprovoked seizure, focal discharge, partial seizure, seizure recurrence, time to recurrence.

A first unprovoked seizure may represent either the onset of epilepsy or an isolated event. Seizure and epilepsy are not synonymous, with epilepsy referring specifically to recurrent, unprovoked seizures.<sup>1,2</sup>

Most children with a single unprovoked seizure do not experience recurrence.<sup>3,4</sup> The issue, however, of whether to treat a patient after a first unprovoked seizure has been the subject of much debate.<sup>5,6</sup> In children, the high incidence of cognitive and behavioral side-effects of anti-epileptic drugs (AED) has led to a growing debate concerning whether long-term pharmacotherapy with AED is justified after a single unprovoked seizure. Knowledge of the history after a single unprovoked seizure and risk factors for recurrence are critical for rational decision making regarding long-term treatment with AED, which is a particularly important issue in children.

The time to recurrence has been reported in various studies. Eighty percent of the 5 year recurrence risk appears to be realized by 2 years after the first unprovoked seizure.<sup>7,8</sup> Few studies,

however, have reported the time to recurrence in detail within 2 years, particularly in Japan. In addition, several predictive factors of recurrence have been suggested, including age at onset, sex, perinatal and family histories, seizure characteristics, whether the child was awake or asleep at the time of seizure, and epileptiform patterns on electroencephalogram (EEG).<sup>9,10</sup> No consensus has been reached, however, especially in children without brain magnetic resonance imaging (MRI) abnormalities. A recent meta-analysis showed that many of the differences between study findings can be attributed to differences in methods and the distribution of important risk factors in the population studied.<sup>7</sup> In particular, EEG helps to establish the diagnosis of epilepsy and define the epileptic syndrome. In patients with a single unprovoked seizure, however, EEG data are scarce.

The literature on seizure recurrence in pediatric patients is highly variable, depending on the follow-up period and the population studied. The present prospective study was undertaken to clarify the time to recurrence and risk factors for seizure recurrence following a first unprovoked seizure in Japanese children.

## Methods

Participants were recruited from University of Yamanashi Hospital between 1 July 1997, and 30 June 2009. The records of each

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participant were reviewed to document the occurrence of a seizure, classify the cause, and verify residency. Eligible candidates were children between 1 month and 15 years old who presented with their first unprovoked afebrile seizure. Participants with multiple seizures occurring within 24 h were included, if they were not taking AED. Participants with previous neonatal seizures, febrile convulsions, immediate post-traumatic seizures, or other provoked seizures who now presented with a first unprovoked seizure were also included. Exclusion criteria were ambiguous episodes, seizures occurring with acute encephalopathy (e.g. meningitis, encephalitis, or electrolyte or glucose disturbances, within 1 week of significant head trauma, or in association with febrile illness), or presence of organic brain lesions on MRI, such as tumor, brain injuries, progressive neurological diseases, or metabolic diseases.

In addition, typical absence, myoclonic, and infantile spasms are very unlikely to present as a first seizure.<sup>11</sup> Accordingly, children with these seizures were excluded because they never present with a single event. Furthermore, children with status epilepticus (defined as a seizure lasting >30 min or a series of seizures lasting >30 min without the child regaining consciousness) were also excluded because of the need for AED.

A seizure was considered unprovoked when no identifiable proximate insult (e.g. fever or head trauma) was apparent that could account for the seizure.<sup>10</sup> A family history of seizures was defined as a history of unprovoked seizure or seizures in a first-degree relative. Seizures were classified as generalized tonic or tonic-clonic, atonic, simple partial, complex partial, or partial secondarily generalized in accordance with International League Against Epilepsy (ILAE) criteria.<sup>12</sup>

At the time of the initial visit, informed consent was obtained from the parents, and assent from the child was obtained when appropriate. Detailed data were collected and coded regarding the exact nature of the seizure, including seizure characteristics, duration, number of seizures in the first 24 h, and any treatment given. Additional information regarding prior provoked seizures, prior neurological insults, birth history, and family history was also coded. All children underwent physical and neurological examinations. Neurological examination was coded as abnormal when any abnormal neurological sign was identified. Further laboratory examinations were performed when clinically indicated.

EEG was performed for all subjects using a 10- or 20-channel machine with the 10–20 electrode system. All records were interictal, obtained >1 week after the seizure. When feasible, both awake and asleep EEG tracings were obtained. EEG reports were reviewed by one of three authors (S.M., H.K. or F.S.) at their respective sites. Although we made no formal study of interrater reliability, all difficult or borderline EEG were reviewed by at least two interpreters. EEG were classified for the presence or absence of overall abnormal interpretation, epileptiform discharges, and/or slowing. Epileptiform abnormalities, including focal spikes, multifocal spikes, rolandic spikes, generalized spike-and-wave complexes, and photoparoxysmal responses, as well as focal and generalized slowing, were each coded separately.

After enrollment, patients were monitored by means of regular visits or telephone interviews at least every 3 months, in which we investigated recurrence of seizures, interval between first and second seizures, clinical events, and introduction and control of seizures with AED. Recurrence was defined as any unprovoked seizure occurring >24 h after the first seizure. This study had two goals: (i) estimate the time to recurrence after a first seizure; and (ii) determine which factors identify individuals at higher and lower risk of experiencing a recurrent seizure and quantify those risks. All subjects were followed for at least 2 years.

### Statistical analysis

Statistical analysis was performed using SPSS II (IBM, Chicago, IL, USA) for descriptive analyses: mean, median, mode, and standard deviation. We used log-rank for bivariate analysis to check associations and hazard ratios (HR) of variables, and clinical outcome (recurrence) during follow up. Statistical significance was established at the 5% level ( $P < 0.05$ ).

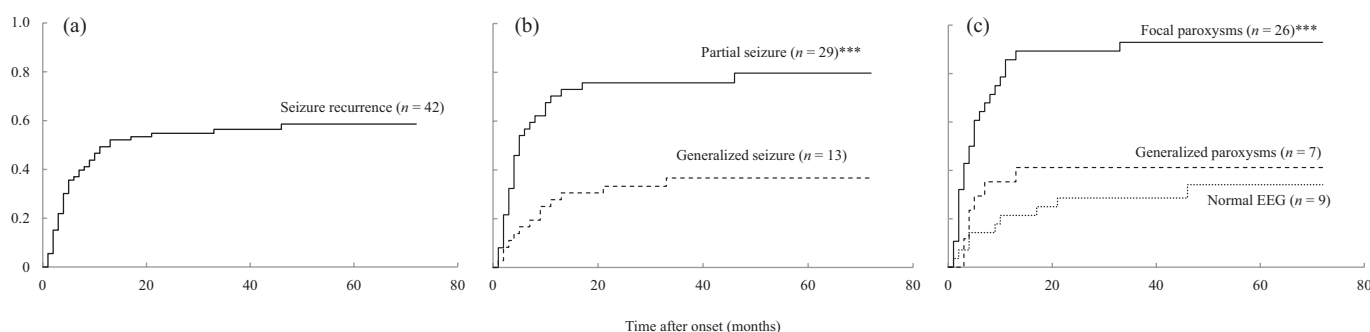
This study was approved by an internal review board. Informed consent was obtained from the patients and the parents of each patient.

### Results

A total of 81 children met all entry criteria and underwent EEG. Eight were excluded because no follow up was obtained after first unprovoked seizure. Thus data were analyzed for 73 children. Candidates were children between 3 months and 13 years old (mean age, 4.3 years; infants,  $n = 12$ ; toddlers,  $n = 47$ ; school children,  $n = 14$ ). Mean duration of follow up after the first seizure was 4.6 years, with a maximum of 6 years.

Of 73 subjects, 42 (57.5%) had recurrence. The overall product-limit estimate of recurrence was 61.9% at 6 months, 85.7% at 1 year, and 95.2% at 2 years after onset of seizure (Fig. 1a). No recurrence was evident >4 years after the first seizure. The median time to recurrence was 7.3 months (range, 2 months–3.9 years). Of the 42 recurrences, 40 (95.2%) occurred within 2 years of the initial seizure, and 42 (100%) occurred within 4 years. Only two (6.0%) of the 33 children who remained seizure free for 2 years after the initial seizure, had seizure recurrence. For the 33 children who were seizure free for 2 years after the initial seizure, risks of late recurrence were 5.9% and 11.8% at 3 and 4 years after the initial seizure, respectively.

Neurological development was normal in 66 patients (90.4%). All patients with developmental delay had mild psychomotor developmental delay. In these patients, only two patients had recurrence. Reports of events in the perinatal period were made in eight cases (11.0%): risk of miscarriage, premature delivery and maternal hypertension. Twelve (16.4%) of the 73 patients had a family history of seizure. Of the 12 patients with family history, eight (66.7%) had recurrence. Table 1 summarizes neurodevelopmental factors, perinatal events, and family history of seizures, as well as HR for seizure recurrence. Overall, risk of recurrence was increased among patients with focal spikes and partial seizures (Table 1). We observed abnormalities in nine patients (12.3%) on neurological examination, including abnor-



**Fig. 1** Kaplan–Meier curves for probability of (a) seizure recurrence after a first unprovoked seizure; and (b,c) seizure recurrence after a first unprovoked seizure as a function of (b) seizure type and (c) electroencephalogram (EEG) paroxysm. (b) Compared with the generalized seizure group, the probability of recurrence for the partial seizure group was 56.8% (hazard ratio [HR], 3.06; 95%CI: 1.46–6.42; \*\*\* $P < 0.001$ ). (c) Compared with normal EEG, the probability of recurrence for the focal paroxysm group was 64.3% (HR, 8.92; 95%CI: 3.41–23.36; \*\*\* $P < 0.001$ ). In contrast, that for the generalized paroxysm group was 29.4% (HR, 1.49; 95%CI: 0.55–4.04).

mal muscle tone (hypotonia) and Todd's paresis. In these patients, one of six patients with hypotonia and one of three patients with Todd's paresis had seizure recurrence. These abnormalities were not associated with seizure recurrence. Overall, no association was identified between history of prior febrile seizure and recurrence. Eight (66.7%) of 12 patients with prior history of simple febrile seizures experienced recurrence, compared with 34 (55.7%) of 61 children with no prior febrile seizures.

Recurrence with partial and generalized seizures occurred in 29 patients (69.0%) and in 13 (31.0%), respectively. Partial seizures were distributed as follows: simple,  $n = 4$  (13.8%); complex,  $n = 14$  (48.3%); and with secondary generalization,  $n = 11$  (37.9%). Generalized seizures were classified as tonic–clonic ( $n = 7$ , 53.8%), tonic ( $n = 2$ , 15.4%), or clonic ( $n = 4$ , 30.8%). Most seizures were short, lasting up to 5 min in 37 patients (88.1%). Most children ( $n = 28$ , 66.7%) had the first seizure while awake. Children with partial seizures experienced recurrence significantly more often than those with generalized seizures ( $P < 0.001$ ; Fig. 1b). No children were given AED after the first seizure.

Normal findings on EEG were followed by recurrent seizures in nine of 42 (21.4%), generalized spike-wave in seven of 42 (16.7%), and focal epileptic discharge in 26 of 42 (61.9%; Fig. 1c). The locations of EEG focal abnormalities with recurrence were as follows: frontal,  $n = 15$  (57.7%); centrottemporal,  $n = 10$  (38.5%); and occipital,  $n = 1$  (3.8%). Children with focal

epileptic discharge had recurrence significantly more often than children with normal findings on EEG ( $P < 0.001$ ), whereas children with generalized spike-and-wave complexes were not significantly different from those with normal findings (Fig. 1c). The relationship between the classification of epilepsy according to ILAE and seizure recurrence is summarized in Table 2.<sup>13</sup>

## Discussion

In this prospective study, we investigated seizure recurrence rates over the course of 2–6 years in children who had experienced a single seizure, and the utility of EEG findings in predicting recurrence. Risk of recurrence after a first unprovoked seizure within 2–6 years of follow up was 57.5%, which is similar to rates reported in two other prospective studies that included adults.<sup>3,10</sup> In addition, the time to seizure recurrence was within 1 year (85.7%), and in particular within 6 months (61.9%) after the first unprovoked seizure. Further, in the present study, focal spikes on EEG predicted a higher rate of recurrence than normal EEG. The

**Table 2** Classification of epilepsy with seizure recurrence

|                                                                       | <i>n</i> | %    |
|-----------------------------------------------------------------------|----------|------|
| Localization-related epilepsy                                         | 35       | 83.3 |
| Idiopathic epilepsy                                                   |          |      |
| Benign childhood epilepsy with centro-temporal spikes                 | 11       | 26.2 |
| Panayiotopoulos syndrome                                              | 4        | 9.5  |
| Benign partial epilepsy with complex partial seizure                  | 2        | 4.8  |
| Symptomatic or non-idiopathic                                         |          |      |
| Frontal lobe epilepsy                                                 | 8        | 19.0 |
| With complex partial seizures, unknown localization                   | 1        | 2.4  |
| With partial seizures, secondary generalization, unknown localization | 9        | 21.4 |
| Generalized epilepsy                                                  | 6        | 14.3 |
| Non-idiopathic                                                        |          |      |
| Generalized epilepsy with tonic–clonic seizures                       | 6        | 14.3 |
| Undetermined                                                          | 1        | 2.4  |
| Total                                                                 | 42       | 100  |

**Table 1** Risk factors for seizure recurrence

| Risk factor (no. subjects)              | HR    | 95%CI        | <i>P</i> |
|-----------------------------------------|-------|--------------|----------|
| Male ( $n = 36$ )                       | 0.849 | 0.439–1.641  | 0.63     |
| Perinatal events ( $n = 8$ )            | 0.592 | 0.171–2.052  | 0.41     |
| Family history of seizures ( $n = 12$ ) | 0.608 | 0.259–1.432  | 0.26     |
| Developmental delay ( $n = 7$ )         | 0.244 | 0.070–0.854  | 0.03     |
| Neurological findings ( $n = 9$ )       | 0.231 | 0.053–1.015  | 0.05     |
| Partial seizures ( $n = 37$ )           | 3.057 | 1.455–6.424  | <0.001   |
| Abnormal EEG ( $n = 45$ )               | 3.199 | 1.466–6.977  | 0.04     |
| Generalized ( $n = 17$ )                | 1.489 | 0.550–4.036  | 0.43     |
| Focal ( $n = 28$ )                      | 8.920 | 3.405–23.364 | <0.001   |

EEG, electroencephalogram.

present study confirmed an increased risk of seizure recurrence when the EEG was abnormal, particularly when focal paroxysmal abnormalities were present.

All long-term observational studies of first seizures show that the risk of recurrence is highest during the period immediately after the initial seizure: 80–90% of the patients who have recurrence, experience this within 2 years of the initial seizure.<sup>12</sup> Most recurrences occur early, with approximately 50% of recurrences occurring within 6 months of the initial seizure.<sup>7</sup> Few studies, however, have described the course of seizure recurrence within 2 years in detail, particularly in Japan. The present study does describe the course of seizure recurrence within 2 years, and is in agreement with a study by Berg and Shinnar.<sup>7</sup> Children who have a first unprovoked seizure should be closely followed within the first year, and particularly within the first 6 months after the seizure.

A first seizure in life does not necessarily mean that epilepsy will develop. After two seizures, however, the risk of subsequent seizures increases to  $\geq 70\%$ .<sup>2</sup> A previous study showed that, in febrile seizures, the duration of recognized fever appears to provide useful information about the risk of later development of epilepsy.<sup>14</sup> The risk of seizure recurrence in a child with a first seizure of remote symptomatic etiology or cryptogenic/idiopathic etiology and epileptiform EEG is similar to the risk of recurrence after a second seizure.<sup>7,10,15,16</sup> Therefore, if a drug trial is being designed for new-onset seizures, it may be rational to include children with one seizure and additional risk factors for recurrence.

Three studies used univariate analysis to show that partial seizures were associated with higher risk of recurrent seizures compared with generalized seizures.<sup>1,9,17</sup> In contrast, another study found no significant association between risk of recurrence and seizure classification.<sup>18</sup> In previous studies, partial seizures were consistently associated with increased risk of recurrence regardless of etiology or the presence of EEG abnormalities.<sup>1,19,20</sup> In addition, patients with partial febrile convulsions had a tendency toward higher risk of developing epilepsy.<sup>21</sup> Another previous study showed that patients with febrile seizure presenting with focal (frontal) paroxysmal EEG abnormalities may be at risk for epilepsy.<sup>22</sup> The present results suggest that partial seizures in combination with focal paroxysmal EEG abnormalities are associated with an increased risk of recurrent seizures and are consistent with these findings. In contrast, generalized seizures are easily recognized, but a brief partial seizure can be less impressive and more difficult to identify. Eliciting detailed information from both parents and teenagers on first diagnosis of seizure is thus very important, to determine whether there has been any prior seizure. Longer follow up and further prospective electroclinical studies are necessary to confirm and define the nosological place of this entity.

Abnormal EEG was associated with increased risk for recurrence in children with unprovoked first seizure in both the present study and in other investigations. Winckler and Rotta confirmed an increased risk for seizure recurrence mainly when there were focal paroxysmal abnormalities.<sup>20</sup> The present results support the conclusion that in children with an unprovoked first seizure,

epileptiform EEG abnormalities, particularly focal paroxysmal discharge, are associated with a higher risk of recurrence than normal findings. In contrast, generalized spike-and-wave discharges tended to show a higher rate of recurrence than normal findings, but the difference was not significant. These results may be related to the small sample size. Nonetheless, generalized spike-and-wave discharge is more common than focal spikes in normal children without seizures and may imply less severe brain disturbances.<sup>23</sup> Even with epileptiform abnormalities, however, the risk of recurrence does not exceed 85% after a mean follow up of 6 years. The cumulative risk of developing epilepsy was reduced to 32.1% when EEG findings were normal, representing a lower rate compared with findings from previous reports.<sup>2,24,25</sup>

Significant risk factors for seizure recurrence were partial seizure and focal EEG paroxysms. Abnormal EEG were more common in children with partial seizures than in children with generalized seizures. On multivariate analysis only abnormal EEG with focal epileptic discharge was a significant prognostic factor, with a cumulative risk that was sixfold higher than EEG with generalized paroxysmal abnormalities. The efficacy of treatment with AED after a first seizure must be clarified before treatment can be decided. These data provide information regarding recurrence risk and long-term outcome, which is necessary for making informed treatment decisions. In this study, developmental delay was associated with lower recurrence risk, and abnormal neurological findings prior to seizure were not associated with the risk of recurrence. Further investigations are necessary to confirm these issues as risk factors or not.

## Conclusion

The period of seizure recurrence may be within 1 year, especially within 6 months after a first unprovoked seizure; and partial seizure and abnormal EEG with focal epileptic discharge may be risk factors for seizure recurrence. Further investigations are needed to clarify the utility of treatment with AED after a first unprovoked seizure.

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